In [1]:
```
import pandas as pd

# Sample data
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [10000, 12000, 15000, 13000, 17000, 16000],
    "Profit": [2000, 3000, 4000, 2500, 3500, 3000]
}

df = pd.DataFrame(data)
```

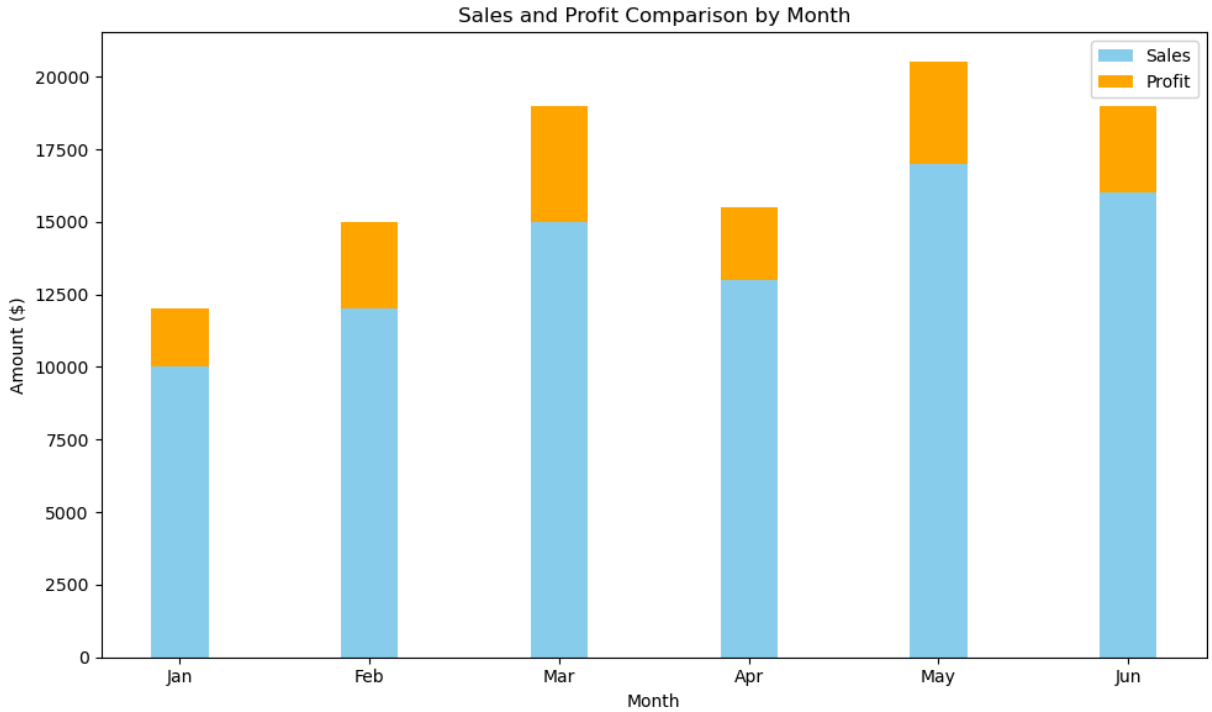
In [2]:
```
df
```

Out[2]:

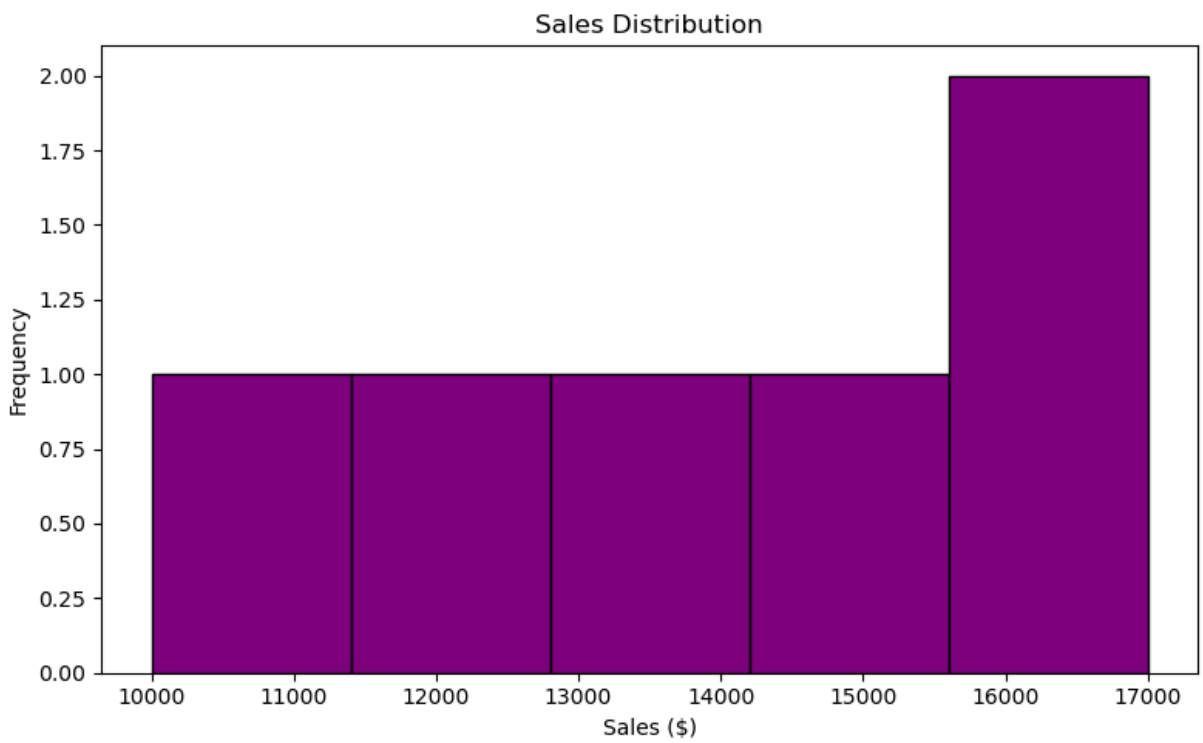
	Month	Sales	Profit
0	Jan	10000	2000
1	Feb	12000	3000
2	Mar	15000	4000
3	Apr	13000	2500
4	May	17000	3500
5	Jun	16000	3000

In [3]:
```
# 1- line plot sales over time

import matplotlib.pyplot as plt

# Line Plot: Sales over Time
plt.figure(figsize=(8, 5))
plt.plot(df['Month'], df['Sales'], color='blue', marker='o', linestyle='-', label='
plt.title('Sales Trend Over Months')
plt.xlabel('Month')
plt.ylabel('Sales ($)')
plt.grid(True)
plt.legend()
plt.tight_layout()
plt.show()
```

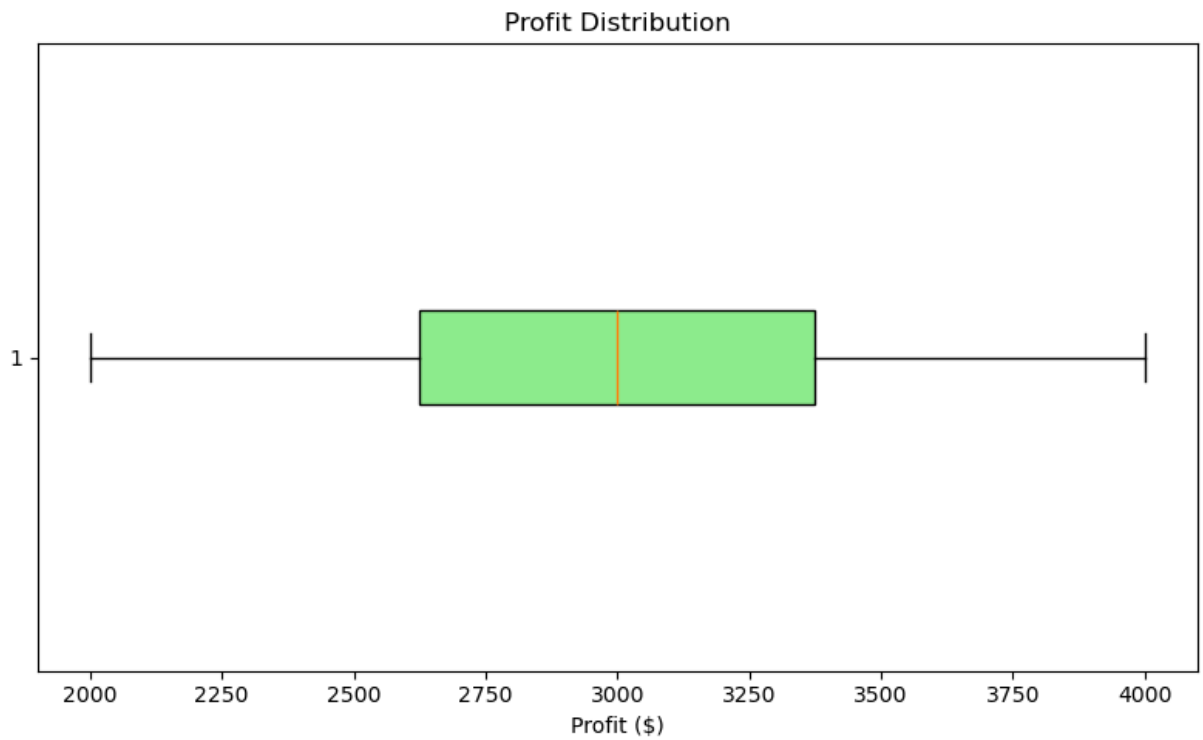
In [4]:
```
# Bar Plot: Sales vs Profit by Month
plt.figure(figsize=(10, 6))
width = 0.3
plt.bar(df['Month'], df['Sales'], width=width, label='Sales', color='skyblue')
plt.bar(df['Month'], df['Profit'], width=width, label='Profit', color='orange', bot
plt.title('Sales and Profit Comparison by Month')
plt.xlabel('Month')
plt.ylabel('Amount ($)')
plt.legend()
plt.tight_layout()
plt.show()

# A stacked bar plot where you can compare Sales and Profit for each month
```



In [5]:
```
# Histogram: Distribution of Sales
plt.figure(figsize=(8, 5))
plt.hist(df['Sales'], bins=5, color='purple', edgecolor='black')
plt.title('Sales Distribution')
plt.xlabel('Sales ($)')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()

# A histogram showing the distribution of Sales values across months.
```



In [6]:
```
# Box Plot: Profit Distribution
plt.figure(figsize=(8, 5))
plt.boxplot(df['Profit'], vert=False, patch_artist=True, boxprops=dict(facecolor="l
plt.title('Profit Distribution')
plt.xlabel('Profit ($)')
plt.tight_layout()
plt.show()

# A box plot showing the spread and outliers of Profit.
```



In [7]:
```
!pip install gradio
```

```
Requirement already satisfied: gradio in c:\users\admin\anaconda3\lib\site-packages
(6.11.0)
Requirement already satisfied: aiofiles<25.0,>=22.0 in c:\users\admin\anaconda3\lib
\site-packages (from gradio) (24.1.0)
Requirement already satisfied: anyio<5.0,>=3.0 in c:\users\admin\anaconda3\lib\site-
packages (from gradio) (4.10.0)
Requirement already satisfied: audioop-lts<1.0 in c:\users\admin\anaconda3\lib\site-
packages (from gradio) (0.2.2)
Requirement already satisfied: brotli>=1.1.0 in c:\users\admin\anaconda3\lib\site-pa
ckages (from gradio) (1.2.0)
Requirement already satisfied: fastapi<1.0,>=0.115.2 in c:\users\admin\anaconda3\lib
\site-packages (from gradio) (0.135.3)
Requirement already satisfied: ffmpy in c:\users\admin\anaconda3\lib\site-packages
(from gradio) (1.0.0)
Requirement already satisfied: gradio-client==2.4.0 in c:\users\admin\anaconda3\lib
\site-packages (from gradio) (2.4.0)
Requirement already satisfied: groovy~=0.1 in c:\users\admin\anaconda3\lib\site-pack
ages (from gradio) (0.1.2)
Requirement already satisfied: hf-gradio<1.0,>=0.3.0 in c:\users\admin\anaconda3\lib
\site-packages (from gradio) (0.3.0)
Requirement already satisfied: httpx<1.0,>=0.24.1 in c:\users\admin\anaconda3\lib\si
te-packages (from gradio) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0,>=0.33.5 in c:\users\admin\anacon
da3\lib\site-packages (from gradio) (1.10.1)
Requirement already satisfied: jinja2<4.0 in c:\users\admin\anaconda3\lib\site-packa
ges (from gradio) (3.1.6)
Requirement already satisfied: markupsafe<4.0,>=2.0 in c:\users\admin\anaconda3\lib
\site-packages (from gradio) (3.0.2)
Requirement already satisfied: numpy<3.0,>=1.0 in c:\users\admin\anaconda3\lib\site-
packages (from gradio) (2.3.5)
Requirement already satisfied: orjson~=3.0 in c:\users\admin\anaconda3\lib\site-pack
ages (from gradio) (3.11.8)
Requirement already satisfied: packaging in c:\users\admin\anaconda3\lib\site-packag
es (from gradio) (25.0)
Requirement already satisfied: pandas<4.0,>=1.0 in c:\users\admin\anaconda3\lib\site
-packages (from gradio) (2.3.3)
Requirement already satisfied: pillow<13.0,>=8.0 in c:\users\admin\anaconda3\lib\sit
e-packages (from gradio) (12.0.0)
Requirement already satisfied: pydantic<=3.0,>=2.0 in c:\users\admin\anaconda3\lib\s
ite-packages (from gradio) (2.12.4)
Requirement already satisfied: pydub in c:\users\admin\anaconda3\lib\site-packages
(from gradio) (0.25.1)
Requirement already satisfied: python-multipart>=0.0.18 in c:\users\admin\anaconda3
\lib\site-packages (from gradio) (0.0.24)
Requirement already satisfied: pytz>=2017.2 in c:\users\admin\anaconda3\lib\site-pac
kages (from gradio) (2025.2)
Requirement already satisfied: pyyaml<7.0,>=5.0 in c:\users\admin\anaconda3\lib\site
-packages (from gradio) (6.0.3)
Requirement already satisfied: safehttpx<0.2.0,>=0.1.7 in c:\users\admin\anaconda3\l
ib\site-packages (from gradio) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in c:\users\admin\anaconda3\lib
\site-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<2.0,>=0.40.0 in c:\users\admin\anaconda3\li
b\site-packages (from gradio) (1.0.0)
Requirement already satisfied: tomlkit<0.14.0,>=0.12.0 in c:\users\admin\anaconda3\l
ib\site-packages (from gradio) (0.13.3)
```

```
Requirement already satisfied: typer<1.0,>=0.12 in c:\users\admin\anaconda3\lib\site
-packages (from gradio) (0.20.0)
Requirement already satisfied: typing-extensions~=4.0 in c:\users\admin\anaconda3\li
b\site-packages (from gradio) (4.15.0)
Requirement already satisfied: uvicorn>=0.14.0 in c:\users\admin\anaconda3\lib\site-
packages (from gradio) (0.44.0)
Requirement already satisfied: fsspec in c:\users\admin\anaconda3\lib\site-packages
(from gradio-client==2.4.0->gradio) (2025.10.0)
Requirement already satisfied: idna>=2.8 in c:\users\admin\anaconda3\lib\site-packag
es (from anyio<5.0,>=3.0->gradio) (3.11)
Requirement already satisfied: sniffio>=1.1 in c:\users\admin\anaconda3\lib\site-pac
kages (from anyio<5.0,>=3.0->gradio) (1.3.0)
Requirement already satisfied: typing-inspection>=0.4.2 in c:\users\admin\anaconda3
\lib\site-packages (from fastapi<1.0,>=0.115.2->gradio) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in c:\users\admin\anaconda3\lib
\site-packages (from fastapi<1.0,>=0.115.2->gradio) (0.0.4)
Requirement already satisfied: certifi in c:\users\admin\anaconda3\lib\site-packages
(from httpx<1.0,>=0.24.1->gradio) (2026.1.4)
Requirement already satisfied: httpcore==1.* in c:\users\admin\anaconda3\lib\site-pa
ckages (from httpx<1.0,>=0.24.1->gradio) (1.0.9)
Requirement already satisfied: h11>=0.16 in c:\users\admin\anaconda3\lib\site-packag
es (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.16.0)
Requirement already satisfied: filelock>=3.10.0 in c:\users\admin\anaconda3\lib\site
-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (3.20.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in c:\users\admin\anaconda3\lib
\site-packages (from huggingface-hub<2.0,>=0.33.5->gradio) (1.4.3)
Requirement already satisfied: tqdm>=4.42.1 in c:\users\admin\anaconda3\lib\site-pac
kages (from huggingface-hub<2.0,>=0.33.5->gradio) (4.67.1)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\admin\anaconda3\li
b\site-packages (from pandas<4.0,>=1.0->gradio) (2.9.0.post0)
Requirement already satisfied: tzdata>=2022.7 in c:\users\admin\anaconda3\lib\site-p
ackages (from pandas<4.0,>=1.0->gradio) (2025.2)
Requirement already satisfied: annotated-types>=0.6.0 in c:\users\admin\anaconda3\li
b\site-packages (from pydantic<=3.0,>=2.0->gradio) (0.6.0)
Requirement already satisfied: pydantic-core==2.41.5 in c:\users\admin\anaconda3\lib
\site-packages (from pydantic<=3.0,>=2.0->gradio) (2.41.5)
Requirement already satisfied: click>=8.0.0 in c:\users\admin\anaconda3\lib\site-pac
kages (from typer<1.0,>=0.12->gradio) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in c:\users\admin\anaconda3\lib\si
te-packages (from typer<1.0,>=0.12->gradio) (1.5.4)
Requirement already satisfied: rich>=10.11.0 in c:\users\admin\anaconda3\lib\site-pa
ckages (from typer<1.0,>=0.12->gradio) (14.2.0)
Requirement already satisfied: colorama in c:\users\admin\anaconda3\lib\site-package
s (from click>=8.0.0->typer<1.0,>=0.12->gradio) (0.4.6)
Requirement already satisfied: six>=1.5 in c:\users\admin\anaconda3\lib\site-package
s (from python-dateutil>=2.8.2->pandas<4.0,>=1.0->gradio) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\admin\anaconda3\lib
\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\admin\anaconda3\l
ib\site-packages (from rich>=10.11.0->typer<1.0,>=0.12->gradio) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in c:\users\admin\anaconda3\lib\site-packa
ges (from markdown-it-py>=2.2.0->rich>=10.11.0->typer<1.0,>=0.12->gradio) (0.1.2)
```

In [8]:

```python
import gradio as gr
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Sample data
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun"],
    "Sales": [10000, 12000, 15000, 13000, 17000, 16000],
    "Profit": [2000, 3000, 4000, 2500, 3500, 3000]
}
df = pd.DataFrame(data)

# Function to return selected plot
def generate_plot(plot_type):
    fig = plt.figure(figsize=(8, 5))

    if plot_type == "Line Plot":
        plt.plot(df['Month'], df['Sales'], color='blue', marker='o', label='Sales')
        plt.title('Sales Trend Over Months')
        plt.xlabel('Month')
        plt.ylabel('Sales ($)')
        plt.grid(True)
        plt.legend()

    elif plot_type == "Stacked Bar Chart":
        fig.set_size_inches(10, 6)
        width = 0.3
        plt.bar(df['Month'], df['Sales'], width=width, label='Sales', color='skyblu
        plt.bar(df['Month'], df['Profit'], width=width, label='Profit', color='oran
        plt.title('Sales and Profit Comparison by Month')
        plt.xlabel('Month')
        plt.ylabel('Amount ($)')
        plt.legend()

    elif plot_type == "Pie Chart":
        fig.set_size_inches(7, 7)
        plt.pie(df['Profit'], labels=df['Month'], autopct='%1.1f%%', startangle=140
        plt.title('Profit Distribution by Month')

    elif plot_type == "Scatter Plot":
        plt.scatter(df['Sales'], df['Profit'], color='green', s=100, edgecolors='bl
        plt.title('Sales vs Profit')
        plt.xlabel('Sales ($)')
        plt.ylabel('Profit ($)')
        plt.grid(True)

    elif plot_type == "Histogram":
        plt.hist(df['Sales'], bins=5, color='purple', edgecolor='black')
        plt.title('Sales Distribution')
        plt.xlabel('Sales ($)')
        plt.ylabel('Frequency')

    elif plot_type == "Box Plot":
        plt.boxplot(df['Profit'], vert=False, patch_artist=True, boxprops=dict(face
        plt.title('Profit Distribution')
        plt.xlabel('Profit ($)')

    plt.tight_layout()
    return fig
```

```
# Gradio UI
demo = gr.Interface(
    fn=generate_plot,
    inputs=gr.Radio(
        ["Line Plot", "Stacked Bar Chart", "Pie Chart", "Scatter Plot", "Histogram"
        label="Choose Plot Type"
    ),
    outputs=gr.Plot(label="Visualization"),
    title="Sales & Profit Visual Explorer",
    description="Choose a chart type to visualize the data."
)

demo.launch()
```

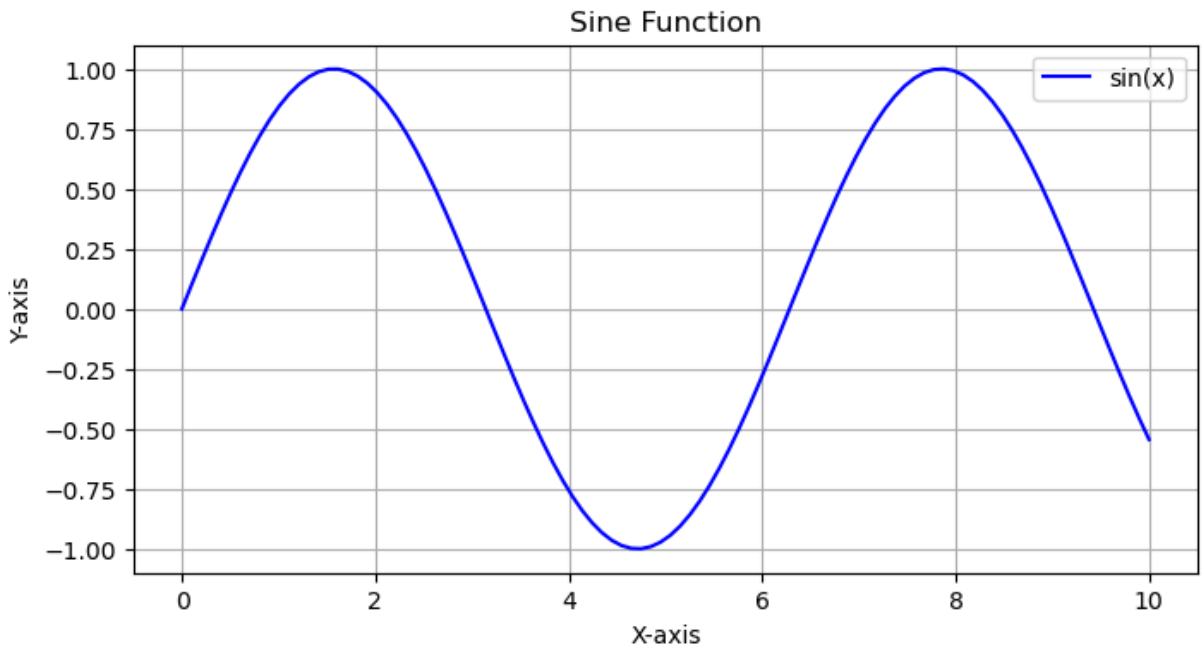
```
* Running on local URL:  http://127.0.0.1:7860
* To create a public link, set `share=True` in `launch()`.
```

Out[8]:

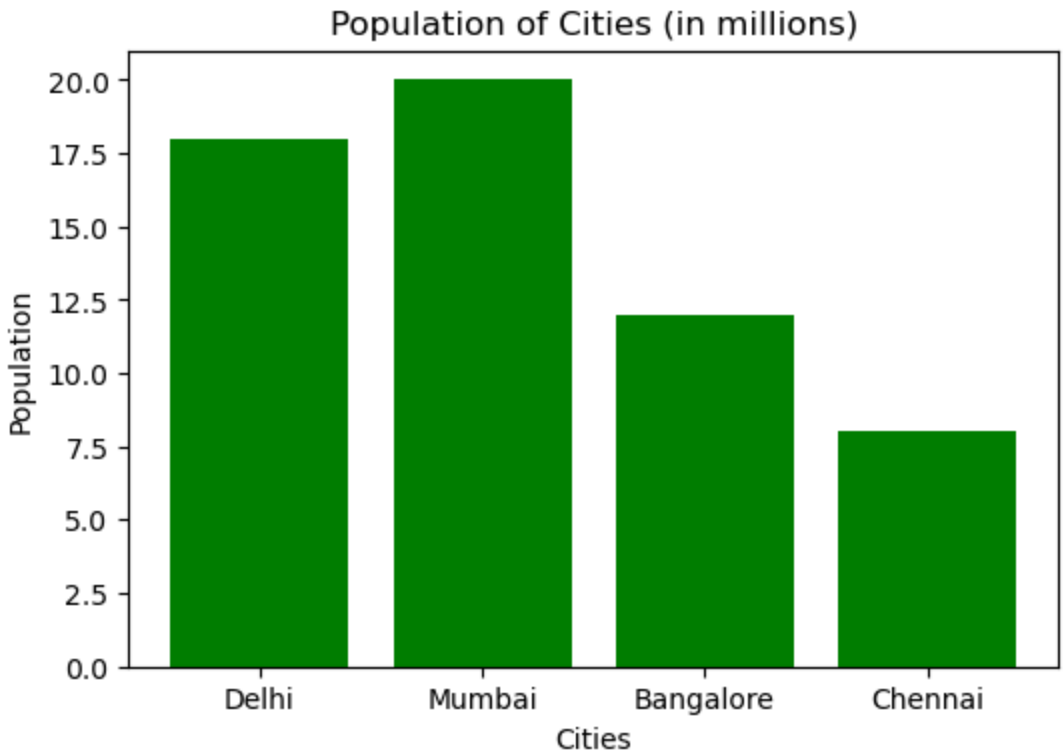
In [9]:
```
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

In [10]:
```
#line plot
x = np.linspace(0, 10, 100)
```

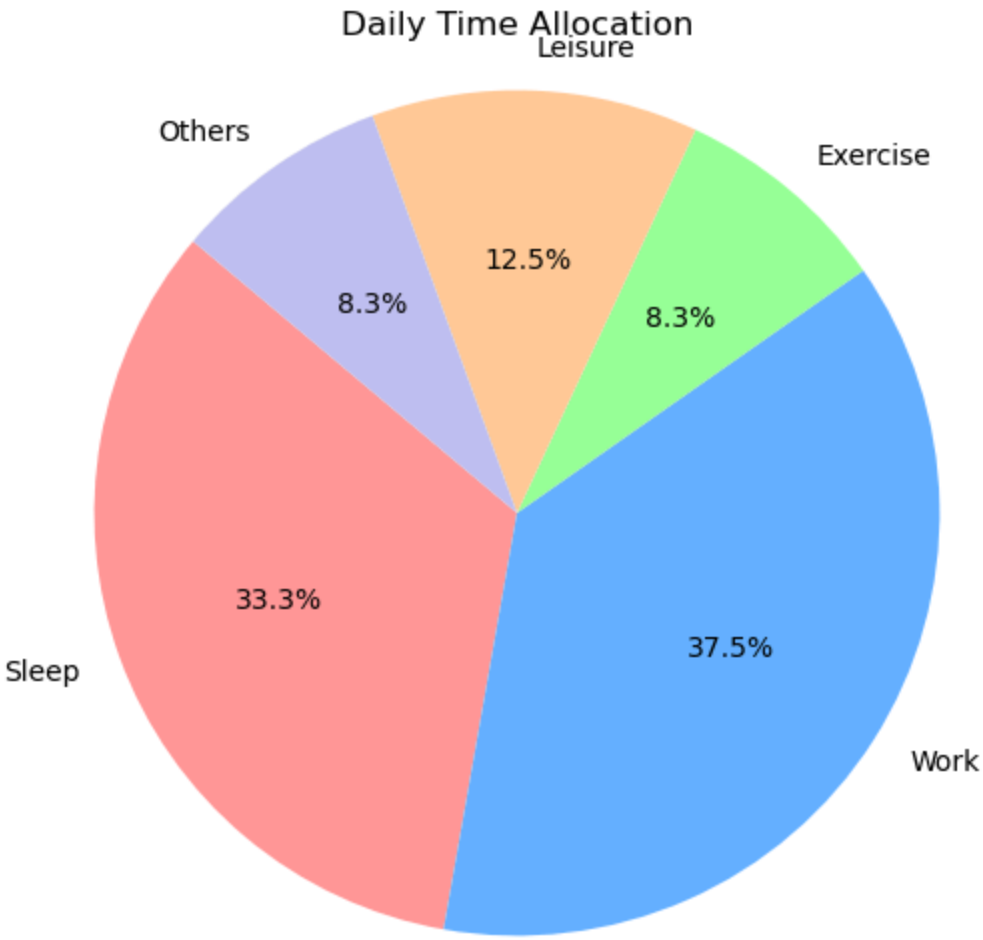
```
y = np.sin(x)

plt.figure(figsize=(8, 4))
plt.plot(x, y, label="sin(x)", color='blue')
plt.title("Sine Function")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.grid(True)
plt.legend()
plt.show()
```



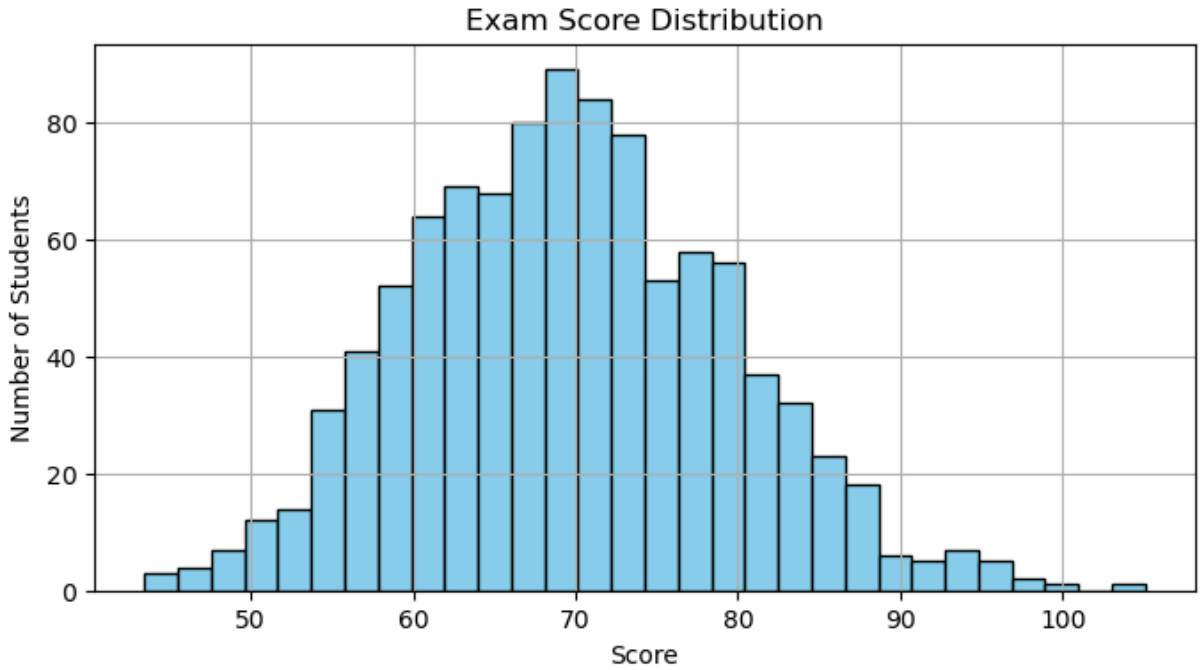
In [11]:
```
#bar chat

cities = ['Delhi', 'Mumbai', 'Bangalore', 'Chennai']
population = [18, 20, 12, 8]

plt.figure(figsize=(6,4))
plt.bar(cities, population, color='green')
plt.title("Population of Cities (in millions)")
plt.xlabel("Cities")
plt.ylabel("Population")
plt.show()
```



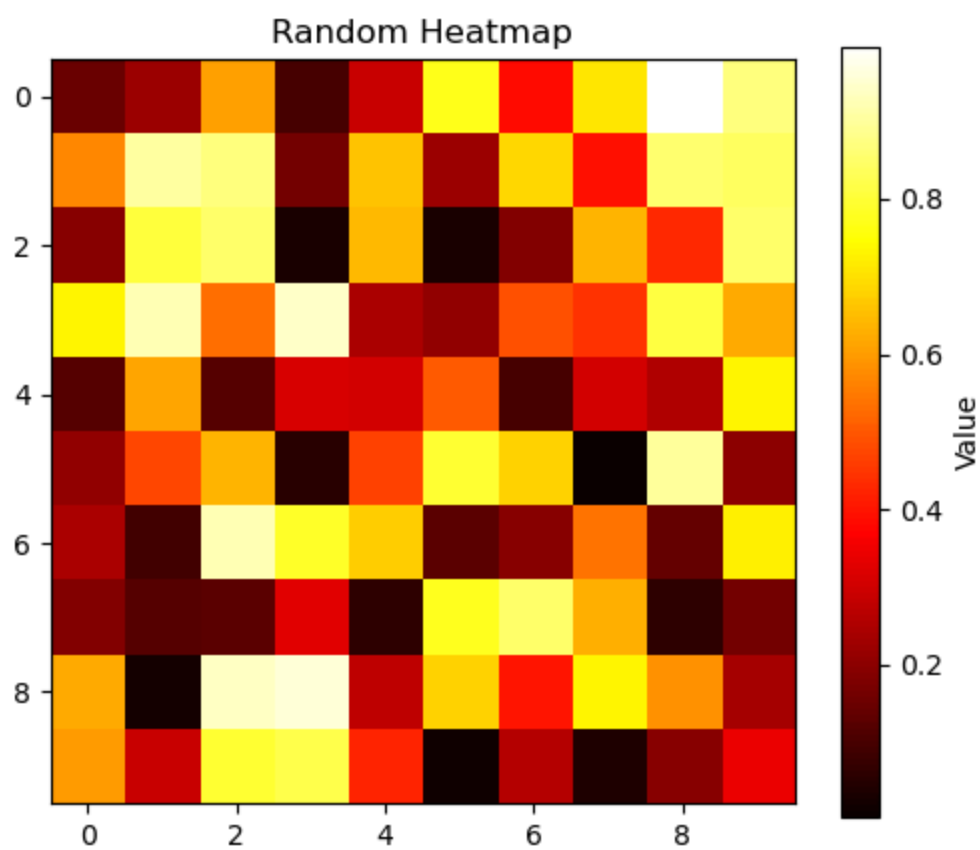
In [12]:
```
#pie chart
labels = ['Sleep', 'Work', 'Exercise', 'Leisure', 'Others']
sizes = [8, 9, 2, 3, 2]
colors = ['#ff9999','#66b3ff','#99ff99','#ffcc99','#c2c2f0']

plt.figure(figsize=(6,6))
plt.pie(sizes, labels=labels, colors=colors, autopct='%1.1f%%', startangle=140)
plt.title("Daily Time Allocation")
plt.axis('equal')  # Equal aspect ratio ensures that pie is drawn as a circle.
plt.show()
```



In [13]:
```
#histogarm
scores = np.random.normal(loc=70, scale=10, size=1000)

plt.figure(figsize=(8,4))
plt.hist(scores, bins=30, color='skyblue', edgecolor='black')
plt.title("Exam Score Distribution")
plt.xlabel("Score")
plt.ylabel("Number of Students")
plt.grid(True)
plt.show()
```



In [14]:
```
#heatmap
data = np.random.rand(10, 10)

plt.figure(figsize=(6, 5))
plt.imshow(data, cmap='hot', interpolation='nearest')
plt.colorbar(label='Value')
plt.title("Random Heatmap")
plt.show()
```



In [ ]:

In [ ]: